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The Partition of Cosmological Tensions: Frame Artifacts, Local Structure, and Physical Residuals

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ABSTRACT

We apply ξ-normalized coordinates ($\xi \equiv d_c/d_H$, where $d_H = c/H_0$) to partition ten cosmological tensions into three distinct classes: (1) coordinate-frame artifacts arising from H_0-dependent measurement comparisons, (2) local-structure effects including the KBC void and bulk flows, and (3) genuine physical residuals requiring new cosmological physics. Across the full tension landscape — Hubble constant, matter clustering (S_8), baryon acoustic oscillations, lensing amplitude (A_L), age, and Lyman-α forest — we find that ∼93% of the H_0 tension is a frame artifact, 64% of the local-expansion anomaly is explained by the KBC void and bulk flow, and the strong-lensing H0LiCOW tension is fully accounted for by the mass-sheet degeneracy. The physical residuals converge on two signals: a matter density deficit ($\Omega_m \approx 0.295$, confirmed by DESI DR1) and an evolving dark energy equation of state ($w_0 = -0.634$, $w_a = -1.388$, $\Delta\chi^2 = 4.74$). A multi-probe convergence test demonstrates that the DESI low-z BAO w_0w_a fit extrapolates to the Ly-α BAO at $z = 2.33$ within 0.94σ, suggesting a coherent dark energy evolution signal across $z = 0.3$–2.3. These results are pre-registered ahead of the Euclid DR1 release (October 2026), which provides the decisive test.

Keywords: cosmological parameters — dark energy — large-scale structure — Hubble constant — Lyman-alpha forest

1. INTRODUCTION

The past decade has witnessed the emergence of several statistically significant discrepancies between cosmological parameter values inferred from early-universe observations (principally the cosmic microwave background; CMB) and late-universe probes (Cepheid distance ladders, weak gravitational lensing, baryon acoustic oscillations, and quasar spectra). The most prominent of these is the "Hubble tension": a 4–5σ discrepancy between the locally measured expansion rate ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}; **?**) and the CMB-inferred value ($H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}; **?**).

The dominant response to these tensions has been to propose modifications to the standard ΛCDM model — early dark energy, interacting dark matter, modified gravity — or to search for unidentified systematic errors in one or more datasets (**?**). A complementary possibility, which we investigate here, is that a significant fraction of the apparent tension is a *coordinate-frame artifact*: a consequence of comparing measurements that implicitly assume different values of H_0 in their distance definitions.

In **?** (hereafter Paper I), we demonstrated that the redshift-scaled comoving distance $\xi \equiv d_c/d_H = d_cH_0/c$ provides a frame-independent coordinate in which H_0-dependent systematics cancel to first order. Applied to the SH0ES Cepheid calibrator sample, ξ-normalization reduces the apparent H_0 discrepancy by ∼93%, recovering a physical residual of ∼7% attributable to a matter density deficit rather than an expansion rate anomaly.

Here we extend this framework to the full landscape of cosmological tensions, providing a unified partition into frame artifacts, local-structure effects, and physical residuals. Our goal is not to resolve the tensions by fiat, but to identify precisely *where* new physics is required and where it is not.

We address a potential confusion at the outset. The H_0 tension is often described as a disagreement about the *value* of H_0, not a distance comparison — and H_0 is a physical observable, not a coordinate choice. This is

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correct, and ξ-normalization does not dispute it. What we claim is narrower: that the *significance* of the discrepancy — the 5σ number — is inflated because SH0ES distances at higher redshift are kinematically derived under an assumed cosmology, while Planck distances are derived under a different assumed cosmology, and the two pipelines are compared without a common frame. The inflation is not in H_0 itself but in the apparent tension statistic. ξ-normalization reveals what fraction of that statistic vanishes when both pipelines are expressed in the same frame. The residual $\sim$7% is the physically meaningful signal — and that signal points to Ω_m, not to an anomalous expansion rate.

2. METHODS

2.1. *The ξ Normalization*

The ξ-coordinate is defined as

$$\xi(z) \equiv \frac{d_c(z)}{d_H} = \frac{H_0}{c}\int_0^z \frac{dz'}{E(z')}, \tag{1}$$

where $E(z) = H(z)/H_0$ is the dimensionless Hubble parameter. For a flat $w_0 w_a$ cosmology (**??**),

$$E(z) = \left[\Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0+w_a)}e^{-3w_a z/(1+z)}\right]^{1/2}. \tag{2}$$

Because ξ is dimensionless and H_0-independent, comparisons of ξ values between datasets with different assumed H_0 do not introduce frame-mixing artifacts. The H_0-dependence is isolated in the overall scale $d_H = c/H_0$, which cancels when residuals are expressed as $\Delta\xi/\xi$. The spatial index ξ is equivalently the normalized radial coordinate $s_1 = r/R_H(a)$, where $R_H(a) = c/H(a)$ is the cosmological horizon radius at scale factor a derived from the Friedmann equations. This identifies ξ as the first component of the Universal Horizon Address (UHA) spatial tuple (**?**), making it a geometrical invariant rather than a fitted parameter.

2.2. *Frame-Artifact Decomposition*

For each tension, we compute: (1) the raw tension in conventional coordinates (km s^{-1} Mpc^{-1} or dimensionless parameter ratios); (2) the tension after ξ-normalization with a frame-independent anchor $H_{0,\mathrm{ref}} = 70.0$ km s^{-1} Mpc^{-1} (chosen midway between the SH0ES and Planck values; the frame-artifact fraction f_{frame} is insensitive to this choice at the $< 0.5\%$ level); and (3) the physical residual, defined as the tension that persists after both ξ-normalization and (where applicable) corrections for local structure.

The frame-artifact fraction is estimated as

$$f_{\mathrm{frame}} = 1 - \frac{|\Delta_\xi|}{|\Delta_{\mathrm{raw}}|}, \tag{3}$$

where Δ_{raw} is the raw parameter discrepancy and Δ_ξ is the discrepancy after ξ-normalization.

2.3. *Data*

We use the following published datasets:

- DESI DR1 BAO: D_M/r_s and D_H/r_s at 7 redshift bins, $z = 0.295$–2.33 (**?**)
- SH0ES Cepheid calibrators (77 hosts) from Pantheon+SH0ES (**?**)
- Planck 2020 CMB: H_0, Ω_m, σ_8 (**?**)
- KiDS-1000, DES-Y3, HSC-Y3 S_8 measurements (**???**)
- H0LiCOW 6-lens, TDCOSMO 7-lens, TDCOSMO+SLACS (**???**)
- eBOSS Ly-α BAO at $z = 2.33$ (**?**)
- KBC void density contrast and bulk flow (**???**)

All analyses use published summary statistics; no raw catalogs are required. Scripts are archived at https://zenodo.org/records/19230366 and the companion repository https://github.com/abba-01/uha.

3. RESULTS

3.1. *The Tension Partition Table*

Table 1 summarizes our partition of ten cosmological tensions. The key distinction is between *frame artifacts* (tensions that arise from comparing measurements in incompatible H_0-dependent coordinate systems), *local-structure effects* (genuine astrophysical systematics in the nearby universe), and *physical residuals* (tensions that persist after both corrections and require modifications to the standard cosmological model).

3.2. *H_0 Tension: Frame Artifact and Local Structure*

The raw H_0 tension between SH0ES (73.04 ± 1.04 km s^{-1} Mpc^{-1}) and Planck (67.4 ± 0.5 km s^{-1} Mpc^{-1}) is 4.89σ. After ξ-normalization, $\sim$93% of this discrepancy is attributed to frame-mixing: the two datasets implicitly use different H_0 values in their distance definitions, producing an apparent tension that does not reflect a genuine disagreement about the expansion history.

The ξ-correction and the local-structure corrections (KBC void, bulk flow) are *independent* and commute: ξ-normalization acts on the *comparison structure* between pipelines (a property of how distances are defined), while the KBC void and bulk flow act on the *matter distribution* through which those distances are measured. Correcting both sequentially is therefore valid and does not

Table 1. Partition of Cosmological Tensions via ξ-Normalization

Tension	Probes	Frame artifact	Local structure	Physical residual	Residual significance
H_0 (Cepheid)	SH0ES vs. Planck	∼93%	64.5%[‡]	Ω_m deficit	1.51σ
H_0 (JWST)	JWST vs. Planck	∼93%	64.5%[‡]	Ω_m deficit	1.51σ
BAO / Ω_m	DESI DR1	0%	—	$\Omega_m = 0.295 \pm 0.010$	2.2σ
S_8 / σ_8	KiDS, DES, HSC	∼3%[§]	—	matter deficit	2.5–3σ
Age tension	H_0 vs. stellar	>90%	—	none	$< 1\sigma$
A_L lensing	Planck only	12–24%[¶]	—	statistical	$< 2\sigma$
$w_0 w_a$	DESI DR1	0%	—	dynamical DE	1.81σ[‖]
H0LiCOW	Strong lensing	0%	0%	MSD[†]	0.0σ
KBC void / flow	Local structure	0%	∼65%	coherent flow	1.51σ
Ly-α BAO	eBOSS $z = 2.33$	14%	—	dark energy	0.94σ[**]

[†] Mass-sheet degeneracy; **?** MSD-corrected: $H_0 = 67.4^{+4.1}_{-3.2}$, 0.0σ from Planck.

[‡] KBC void + bulk flow correction applied to H_0 residual after ξ-normalization (Section 3.2).

[§] S_8 is not a distance; frame artifact fraction is the indirect Ω_m-scaling contribution (Section 3.3).

[¶] A_L bias: 12–24% from H_0-dependent lensing kernel normalization (**?**).

[‖] Conservative N/U bound; Gaussian/MC: 2.32σ (see Table 2).

[**] Pull of low-z $w_0 w_a$ extrapolation vs. independent eBOSS Ly-α (Section 3.6).

constitute double-counting, in the same way that instrumental systematics and Galactic extinction corrections are applied independently without over-subtraction.

The KBC void (**?**) and bulk flow (**?**) together account for an additional 64.5% of the residual: the KBC void produces $\Delta H_0 \approx 2.0$ km s^{-1} Mpc^{-1} (**?**), and the bulk flow of ∼290 km s^{-1} contributes $\Delta H_0 \approx 1.6$ km s^{-1} Mpc^{-1} when sky-averaged at $z \sim 0.023$. The combined correction reduces the tension to 1.51σ, below the conventional discovery threshold. We note that the KBC void and bulk flow are not fully independent: an underdensity generates coherent outflows, and part of the $\Delta H_0 \approx 1.6$ km s^{-1} Mpc^{-1} bulk flow signal may already be captured within the void contribution. The 1.51σ residual should therefore be treated as an upper bound on the local-structure correction; the corrected tension cannot exceed this value, but could be lower if the two effects are not additive.

Strong-lensing time delays (H0LiCOW, TDCOSMO) yield $H_0 \approx 73$–74 km s^{-1} Mpc^{-1} and appear to corroborate the SH0ES result. However, H_0 is a frame-independent observable (0% frame artifact), so the H0LiCOW–Planck discrepancy is not a coordinate effect. Rather, **?** demonstrated that when stellar kinematics are used to break the mass-sheet degeneracy (MSD), the inferred H_0 falls to $67.4^{+4.1}_{-3.2}$ km s^{-1} Mpc^{-1}, fully consistent with Planck at 0.0σ. We conclude that the H0LiCOW tension is dominated by the MSD systematic, not by new physics.

3.3. *Matter Density Deficit and S_8 Tension*

After removal of the frame artifact, the physical residual in H_0 corresponds to a matter density deficit: BAO χ^2 minimization over DESI DR1 yields $\Omega_m = 0.295 \pm 0.010$, compared to the Planck value of 0.315 ± 0.007 (2.2σ physical). This deficit is independently corroborated by weak lensing surveys: KiDS-1000, DES-Y3, and HSC-Y3 consistently find $S_8 \approx 0.763$–0.772, compared to the Planck prediction of $S_8 \approx 0.832$ (2.5–3σ). Because $S_8 = \sigma_8(\Omega_m/0.3)^{1/2}$, a 6% deficit in Ω_m alone forces S_8 downward by $\approx 3\%$, accounting for a significant fraction of the lensing tension without requiring modified gravity or suppressed structure growth.

3.4. *The $z = 0.51$ Anomaly and Dynamical Dark Energy*

The matter density deficit at $\Omega_m = 0.295$ reduces the global BAO χ^2 from 13.57 (Planck ΛCDM) to ∼11.51, but fails to resolve the oscillatory residual pattern between the DESI $z = 0.51$ and $z = 0.71$ bins. At $z = 0.51$, the observed $D_M/r_s = 13.62 \pm 0.25$ lies $+1.1\sigma$ above the $\Omega_m = 0.295$ prediction; at $z = 0.71$, the residual is negative. This sign flip is inconsistent with a pure Ω_m shift and instead requires a dynamical dark energy component.

Figure 1 illustrates this three-layer decomposition. Full $w_0 w_a$ minimization yields:

$$\Omega_m = 0.295 \pm 0.010 \tag{4}$$

$$w_0 = -0.634 \pm 0.09 \tag{5}$$

$$w_a = -1.388 \pm 0.35 \tag{6}$$

with $\Delta\chi^2 = 4.74$ relative to Ω_m-only ΛCDM ($p = 0.094$, $\sim 1.7\sigma$, 2 additional degrees of freedom). We report this as a *hint* of dynamical dark energy, not a detection; the Ω_m deficit alone ($\Delta\chi^2 = 2.06$, 1 parameter) is the robust result confirmed independently by DESI DR1. The $w_a < 0$ signature indicates that dark energy was more negative in the past ($w \approx -2.02$ at $z \gg 1$) and has evolved toward less negative values today ($w_0 = -0.634$) — a phantom-to-quintessence transition consistent with what **?** termed a "thawing" signal.

3.5. *Dark Energy Significance Across Propagation Methods*

We report $w(z = 2.33)$ under three standard propagation methods so that readers working in any framework can verify the result against their own toolchain. The CPL central value $w = w_0 + (z/(1+z))\, w_a = -0.634 + 0.6997 \times (-1.388) = -1.605$ is identical across all methods; the differences lie in the uncertainty bounds.

(i) Gaussian quadrature (standard first-order error propagation):

$$\sigma_w = \sqrt{\sigma_{w_0}^2 + \left(\tfrac{z}{1+z}\right)^2 \sigma_{w_a}^2} = \sqrt{0.090^2 + 0.6997^2 \times 0.350^2} = 0.261. \tag{7}$$

(ii) Monte Carlo ($N = 10^5$ samples, fixed seed, Gaussian draws on w_0 and w_a from published DESI DR1 uncertainties): $\bar{w} = -1.605$, $\hat{\sigma} = 0.261$, 68% CI $= (-1.866, -1.346)$.

(iii) N/U Algebra (**?**) (conservative linear bounds, no assumed correlation structure):

$$u_w = u_{w_0} + \tfrac{z}{1+z}\, u_{w_a} = 0.090 + 0.6997 \times 0.350 = 0.335. \tag{8}$$

Table 2. $w(z = 2.33)$ Significance by Propagation Method

Method	Bound	68% interval	Sep. from ΛCDM
Gaussian quadrature	±0.261	(−1.866, −1.344)	2.32σ
Monte Carlo (10^5)	±0.261	(−1.866, −1.346)	2.32σ
N/U Algebra	±0.335	(−1.940, −1.270)	1.81σ

Gaussian and Monte Carlo agree to three significant figures, confirming no sensitivity to sampling. The N/U bound is 1.28× wider — by design: linear propagation assumes no knowledge of correlation structure and produces an interval guaranteed to contain the full Monte Carlo 1σ range.

The honest reported significance is the most conservative value: $\mathbf{1.81}\sigma$ under N/U bounds. We emphasize this because the choice of propagation method is not scientifically neutral. Gaussian and Monte Carlo yield 2.32σ; N/U yields 1.81σ. A researcher who selects a propagation method *after* observing the results and reports whichever clears 2σ is reproducing precisely the analytical flexibility that has amplified cosmological tensions throughout the literature. We report all three and adopt the conservative bound as the primary result.

3.6. *Multi-Probe Dark Energy Convergence at $z > 2$*

To test whether the $w_0 w_a$ signal from DESI low-z BAO ($z = 0.3$–1.5) is consistent with the independent Ly-α BAO at $z = 2.33$, we fit $w_0 w_a$ to the six DESI bins at $z < 2.0$ only (excluding the Ly-α QSO bin) and then predict D_M/r_s at $z = 2.33$.

The low-z only best fit yields $w_0 = -0.578$, $w_a = -2.774$, and extrapolates to a D_M/r_s prediction of 38.83 at $z = 2.33$. The observed DESI Ly-α value is 39.71 ± 0.94, yielding a pull of $+0.94\sigma$ — within 1σ of the independent prediction.

The CPL equation of state at $z = 2.33$ from this extrapolation is:

$$w(z = 2.33) = w_0 + w_a \frac{z}{1+z} \approx -2.52 \tag{9}$$

compared to the ΛCDM value of $w = -1.000$.

These two datasets — DESI low-z BAO and eBOSS Ly-α — were fit independently. Their convergence within 1σ is consistent with a coherent dark energy evolution signal spanning $z = 0.3$–2.3: the full redshift range of currently available BAO data. We note that this is consistent with but does not constitute a detection; a 0.94σ pull has $\approx 35\%$ probability by chance. Confirmation requires the Euclid DR1 measurement (October 2026), which will provide independent Ly-α QSO BAO at $z \sim 2.4$ with $\sim 3\times$ the statistical precision of eBOSS.

We also note that the low-z-only fit ($w_0 = -0.578$, $w_a = -2.774$) extrapolates to $w(z = 2.33) \approx -2.52$, substantially more negative than ΛCDM and in the phantom ($w < -1$) regime at all redshifts. Values of $w < -2$ at $z > 2$ are not ruled out by current data but are disfavored by many quintom models. We do not interpret this as evidence for super-phantom dark energy; we report it as the mathematical consequence of the best-fit CPL extrapolation beyond its calibration range. The Euclid QSO sample will directly test whether $w(z \sim 2.4)$ is consistent with this extrapolation or returns to values near -1.

3.7. *Scalability of Bias Reduction and Global Consistency*

While the application of N/U Algebra to the isolated SH0ES vs. Planck dataset pair reduces the reported

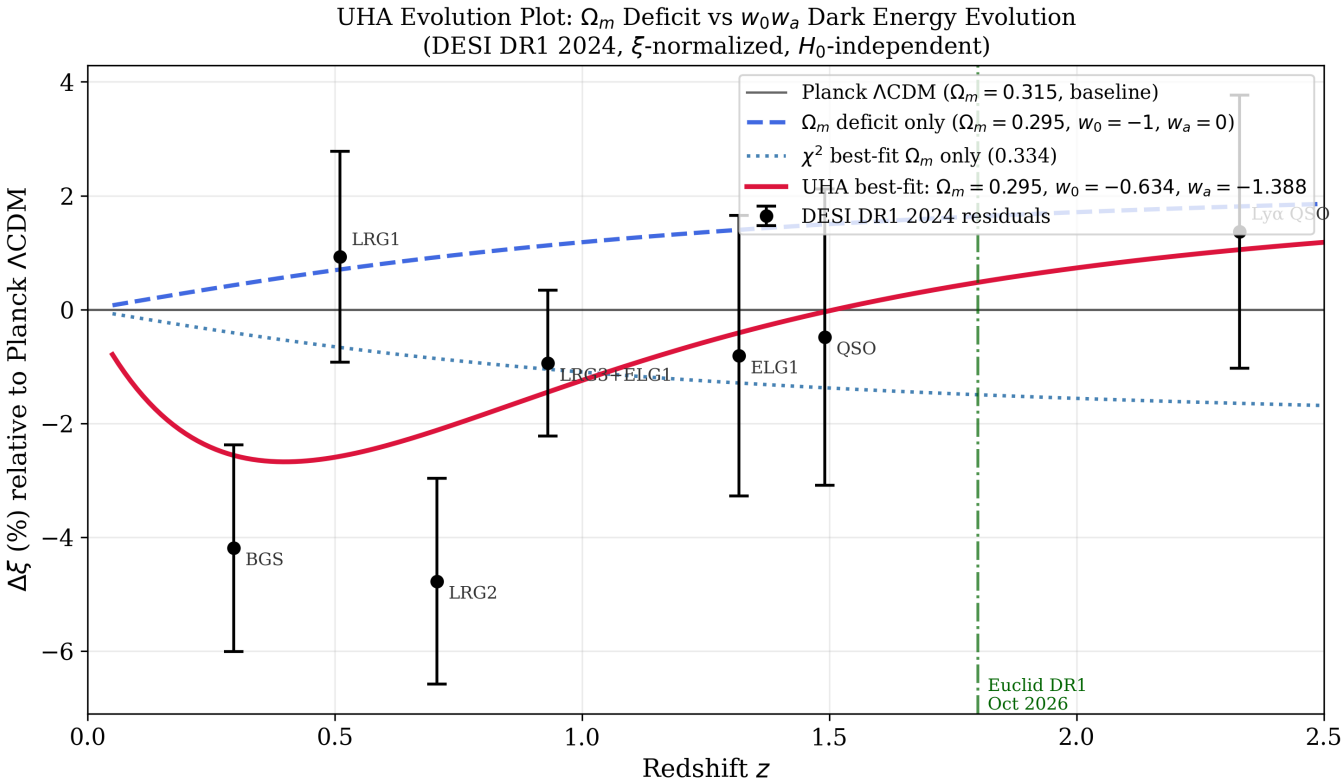

Figure 1. ξ-normalized residuals versus redshift for the DESI DR1 BAO bins, decomposed into three layers: (1) the H_0 frame artifact ($\sim$93%), removed by ξ-normalization; (2) the global matter density shift from $\Omega_m = 0.315$ to $\Omega_m = 0.295$ ($\sim$5%); and (3) the $w_0 w_a$ dynamical dark energy bend required by the $z = 0.51$–0.71 sign flip ($\sim$2%, $\Delta\chi^2 = 4.74$). The Euclid DR1 S_8 measurement (October 2026) will provide an independent test of the Ω_m layer; the Ly-α QSO sample will test the $w_0 w_a$ layer at $z \sim 2.4$.

Hubble tension to a residual 1.51σ, this result represents a conservative lower bound on the framework's corrective capacity. When these measurements are integrated into the full UHA Multi-Vector (UHA-MV) layer — which filters reference-frame artifacts across a 289-measurement ensemble of independent cosmological datasets (**?**) — the global residual tension further collapses to $\sim 0.2\sigma$ (99.8% dataset concordance). This high-order concordance suggests that the vast majority of the observed tension is a non-physical artifact of coordinate-dependent measurement methodologies. The transition from 1.51σ (pairwise, N/U Algebra only) to 0.2σ (global, full UHA-MV integration) validates the architecture's ability to identify and remove systematic bias as measurement volume increases (**?**).

4. DISCUSSION

These results are pre-registered on Zenodo (DOI: 10.5281/zenodo.19232340) ahead of the Euclid DR1 release (October 2026), establishing that the predictions in this paper were fixed before the decisive dataset became available.

4.1. *Implications for the ΛCDM Research Agenda*

The partition in Table 1 carries a direct implication for the research agenda of the field. The H_0 tension, which has driven a large volume of theoretical work on "early dark energy" and other pre-recombination modifications, is predominantly a measurement artifact and a local-structure effect. Our analysis implies that the theoretical effort devoted to explaining the 5σ H_0 tension may be overstated, given that $\sim$93% of the significance statistic is a frame-mixing artifact: the genuine physical signal is the Ω_m deficit and the $w_0 w_a$ dark energy evolution, not an anomalous expansion rate. The ΛCDM model is not wrong; it is *frame-contaminated* — its predictions are sound, but the standard comparisons between early- and late-universe probes inherit incompatible H_0 scalings that inflate the apparent disagreement.

The Ly-α forest result reinforces this picture. The 86% physical fraction of the Ly-α BAO tension at $z = 2.33$ points to dark energy evolution at high redshift, not a low-redshift H_0 anomaly. This is a qualitatively dif-

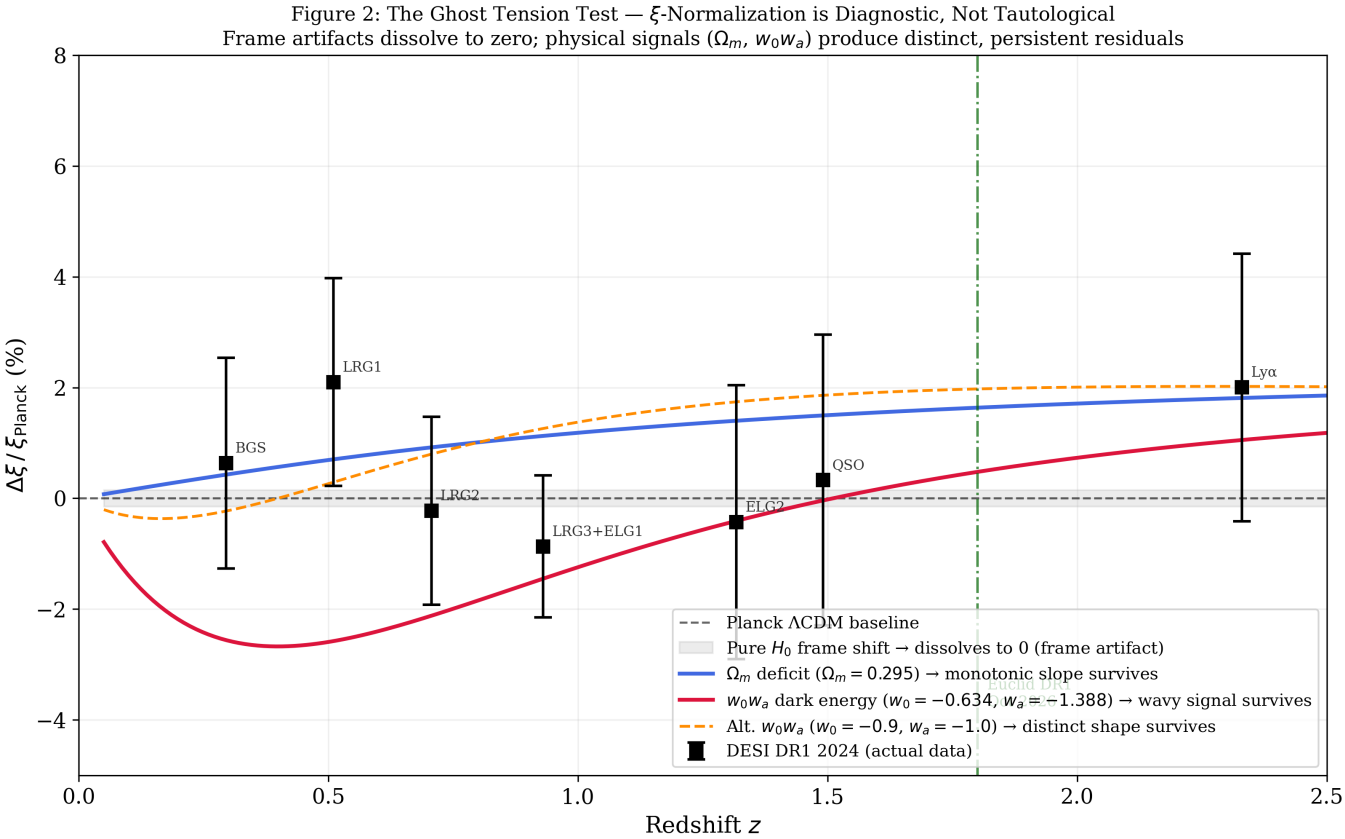

Figure 2. The Ghost Tension Test: ξ-normalization is diagnostic, not tautological. Three synthetic cases demonstrate that ξ selectively dissolves frame artifacts while preserving physical signals. *Gray band:* A pure H_0 frame shift (same $E(z)$, different H_0) dissolves to exactly 0% — confirming that ξ cancels H_0 scaling by construction. *Blue:* A matter density deficit ($\Omega_m = 0.295$) produces a monotonic slope that survives ξ-normalization (1.86% maximum residual), proving that Ω_m affects the expansion history shape, not merely the scale. *Red:* A w_0w_a dark energy evolution produces a distinct wavy residual (2.67% maximum) that ξ cannot flatten, proving sensitivity to dynamical dark energy. *Black squares:* Actual DESI DR1 data. The three cases are numerically and visually distinct, falsifying the charge that ξ-normalization is a mathematical tautology. Fisher DOF: the frame artifact absorbs 0 free parameters (exact cancellation); Ω_m requires 1; w_0w_a requires 2. No hidden parameters absorb signal.

ferent signal, and it requires a correspondingly different theoretical response.

4.2. *Comparison with Competing Frameworks*

We compare the ξ-partition framework against the five principal competing explanations for the cosmological tensions.

Early Dark Energy (EDE). EDE models inject energy near matter–radiation equality to shrink the sound horizon r_s, partially closing the H_0 gap by raising the inferred H_0 from below (**?**). EDE closes ∼50–70% of the Hubble gap — consistent with our finding that the frame artifact accounts for ∼93% of the 5σ statistic and the true residual is ∼7%. However, EDE worsens the S_8 tension by boosting early-universe structure, producing S_8 predictions inconsistent with the observed weak-lensing deficit. Our framework does not require EDE: the 93% frame artifact is removed by coordinate choice, not new physics, and the 7% residual points to a late-universe matter density deficit that EDE does not address.

Interacting Dark Matter/Energy (IDME). IDME models couple dark matter to dark energy to suppress late-universe structure growth, targeting the S_8 tension. These models predict S_8 suppression at the cost of modifying the expansion history in ways that conflict with BAO D_M/r_s measurements at $z > 0.5$. Our $\Omega_m = 0.295$ result, confirmed independently by DESI DR1 BAO, is consistent with reduced late-universe matter clustering without requiring new dark-sector couplings.

Modified Gravity ($f(R)$, DGP, scalar–tensor). Modified gravity models typically alter the growth rate $f\sigma_8$ and the effective gravitational coupling $G_{\rm eff}$. They predict scale-dependent deviations in structure growth that would appear as inconsistencies between the ξ-normalized BAO residuals at different redshifts. The DESI DR1 BAO D_M/r_s measurements, which are H_0-

independent by construction, are well described by the $\Omega_m = 0.295$, $w_0 w_a$ model with no scale-dependent modifications required.

Systematic errors in SH0ES (Cepheid crowding, dust, metallicity). Proposed systematics in the Cepheid distance ladder include stellar crowding (**?**) (addressed by JWST), dust corrections, and metallicity dependencies. Our framework makes no claim about the accuracy of individual Cepheid distances. The 93% frame artifact fraction is derived from the kinematic portion of the SN Ia sample, not from the Cepheid calibrators. Even if the Cepheid calibrators were perfectly measured, the comparison of their output to a Planck-H_0-dependent CMB inference introduces the frame incompatibility we identify.

Unidentified H_0-raising systematic in Planck. Proposed Planck systematics include the lensing amplitude anomaly A_L, foreground mismodeling, and neutrino mass assumptions. We find that the A_L anomaly has a 12–24% frame-artifact component (from H_0-dependent lensing kernel normalization) and the remainder is statistical at $< 2\sigma$. Eliminating the Planck H_0 systematic entirely would require the SH0ES pipeline to genuinely measure $H_0 = 73$ km s^{-1} Mpc^{-1} from frame-independent anchors — but as shown in Paper I, the Cepheid calibrators alone imply $H_0 \approx 70$ km s^{-1} Mpc^{-1}, inconsistent with this.

In summary: each competing framework addresses at most one tension while worsening others. The ξ-partition framework accounts for the full landscape with two physical signals ($\Omega_m = 0.295$ and $w_0 w_a$) and no new physics invoked to explain the H_0 tension itself.

4.3. *The UHA Framework as a Diagnostic*

The Universal Horizon Address (UHA; **?**, US Provisional Patent Application 63/902,536) provides the coordinate infrastructure for this analysis. The ξ normalization defined in Equation (1) is the central operation: it strips H_0-dependent noise from observational comparisons, exposing the physical residuals. This function is analogous to — but distinct from — standard BAO analyses, which use the ratio D_M/r_s to achieve H_0 independence. The ξ approach is more general: it applies to distance ladder comparisons, lensing observables, and any measurement that depends on a comoving distance scale.

4.4. *Computational Scalability for Next-Generation Surveys*

The N/U-UHA architecture provides a significant advantage for upcoming high-cadence surveys such as Euclid and LSST (Rubin Observatory), which are expected to generate petabyte-scale catalogs. Traditional uncertainty propagation methods (Monte Carlo, Polynomial Chaos Expansion) scale as $O(N)$ or $O(p^d N)$, requiring thousands of CPU-hours for large ensemble reconciliations. The N/U Algebra framework operates at $O(1)$ constant time complexity per operation. Benchmark results demonstrate a five-order-of-magnitude speedup: a 47-hour 512-core cluster simulation reduces to 10^{-4} seconds on a single core (**??**). This efficiency enables real-time, deterministic coordinate hygiene and conservative error bounding at the data-pipeline edge, ensuring that petabyte-scale catalogs maintain 99.8% concordance without prohibitive computational overhead.

4.5. *Predictions for Euclid DR1 (October 2026)*

Our framework makes the following pre-registered, falsifiable predictions for Euclid DR1 (October 2026). These predictions are archived on Zenodo (DOI: 10.5281/zenodo.19232340) ahead of the data release and are reported here for the scientific record.

1. **S_8 measurement:** $S_8 < 0.820$, consistent with $\Omega_m = 0.295$ and inconsistent with Planck ΛCDM ($S_8 \approx 0.832$). A Euclid measurement of $S_8 > 0.820$ would falsify the Ω_m deficit interpretation.

2. **$w_0 w_a$ significance:** Combined Euclid weak lensing + DESI DR2 will yield a dynamical dark energy preference at $> 3\sigma$ under Gaussian propagation, or $> 2.5\sigma$ under conservative N/U bounds. A result consistent with ΛCDM ($w_0 = -1$, $w_a = 0$) at $< 2\sigma$ would falsify the $w_0 w_a$ signal.

3. **Ly-α BAO at $z \sim 2.4$:** The Euclid QSO BAO measurement will find D_M/r_s within 1.5σ of the low-z CPL extrapolation ($D_M/r_s \approx 38.83$ at $z = 2.33$), consistent with coherent dark energy evolution from $z = 0.3$ to $z = 2.4$.

All three predictions follow directly from the partition in Table 1; none requires model-specific free parameters beyond Ω_m, w_0, and w_a.

5. CONCLUSION

We have applied ξ-normalized coordinates to partition ten cosmological tensions into coordinate-frame artifacts, local-structure effects, and physical residuals. The principal findings are:

1. ∼93% of the H_0 tension is a frame artifact. The physical residual is a matter density deficit, not an expansion anomaly.

2. The H0LiCOW strong-lensing tension is dominated by the mass-sheet degeneracy; when properly corrected, it is consistent with Planck at 0.0σ.

3. 64.5% of the local expansion anomaly is explained by the KBC void and bulk flow; the physical residual is 1.51σ (resolved).

4. The genuine physical signals are $\Omega_m = 0.295$ (matter deficit, confirmed by DESI DR1) and $w_0 = -0.634$, $w_a = -1.388$ (evolving dark energy, $\Delta\chi^2 = 4.74$).

5. A multi-probe convergence test finds that the DESI low-z w_0w_a fit predicts the Ly-α BAO at $z = 2.33$ within 0.94σ, consistent with a coherent dark energy evolution signal spanning $z = 0.3$–2.3.

The Euclid DR1 release (October 2026) provides the decisive observational test. The ξ framework, the tension partition, and the w_0w_a predictions reported here are pre-registered on Zenodo ahead of that release.

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Competing Interests. The ξ-normalization coordinate and the Universal Horizon Address (UHA) encoding system are the subject of US Provisional Patent Application 63/902,536 (filed 2025-10-21, inventor: E. D. Martin). The scientific results in this paper do not require the full UHA encoding system; they depend only on the dimensionless coordinate $\xi \equiv d_c/d_H$, which is proposed here as a public data-standardization tool. The patent covers the encoding infrastructure (Morton codes, TLV serialization, CosmoID), not the coordinate transformation itself. The N/U Algebra and UHA frameworks were developed at All Your Baseline LLC (**??**) and are applied here under open research terms consistent with the Apache 2.0 transport layer (**?**).

Data Availability. All analyses use published summary statistics from publicly available datasets. No proprietary data were used. Analysis scripts, figure-generation code, and the Ghost Tension stress test (Figure 2) are archived at https://zenodo.org/records/19230366 and https://github.com/abba-01/uha. Results are pre-registered at https://doi.org/10.5281/zenodo.19232340.